

Studying Geographic Generalization in EuroSAT Land-Cover Classification with Multispectral Inputs and Spectral-Group Dropout

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Abstract

Deep learning models for satellite image classification often achieve strong performance on standard i.i.d. benchmarks, but their performance may decline under geographic shift, missing spectral information, or input corruption. This study evaluates geographic, spectral, and noise robustness in EuroSAT land-cover classification using RGB and 13-band multispectral ResNet-50 models. We compare standard random and longitude-based spatial splits, apply spectral-group dropout and noise-augmented training, and perform overlap-controlled class-level cross-evaluation. Geographic sensitivity is concentrated in agricultural classes. The selected $p = 0.5$ dropout model improves robustness to visible and red-edge/NIR masking, but reduces clean macro-F1 and does not improve every masked condition. Noise augmentation improves performance under the two tested corruption types, with some transfer between Gaussian and shot noise. These results show that a single random-split score does not fully describe model robustness. Code and experiment artifacts are available in our GitHub repository.

0.1 Problem Statement and Motivation

The classification of Land-cover from satellite imagery supports environmental monitoring, urban planning, and land-use analysis. EuroSAT is a common benchmark for this task because it contains labeled Sentinel-2 images from ten land-cover classes [1]. However, standard benchmark evaluations often use random data splits. Nearby satellite images can share climate, vegetation, soil, acquisition conditions, and visual structure. Random splits may therefore place spatially related samples in both the training and evaluation sets. Our goal is simple: we test whether a model trained in one part of Europe still works in another region, and whether it remains reliable when some spectral information is missing or noisy.

This does not mean that a model directly memorizes geographic coordinates. However, it can use region-specific correlations that also appear in the random test set. Previous work has shown that standard validation can give optimistic performance estimates when spatial dependence is ignored [2, 3]. A geographically separated evaluation gives a different and often more realistic transfer test.

Geographic separation is only one source of difficulty. Multispectral models also depend on information from sev-

eral wavelength ranges, and their behavior may change when spectral bands are missing or when the input is corrupted by sensor noise. Therefore, this work studies three aspects of model robustness: *geographic generalization* across spatially separated regions, *spectral robustness* to missing band groups, and *noise robustness* to physically motivated input corruptions. We compare standard and spatial splits, RGB and multispectral inputs, spectral-group dropout, noise-augmented training, and class-level cross-evaluation on overlap-controlled test samples.

0.2 Related Work and Current Limits

The original EuroSAT study evaluated pretrained ResNet-50 and GoogLeNet models and compared individual spectral bands and band combinations, demonstrating the effectiveness of CNNs and multispectral information for land-cover classification [1]. TorchGeo later provided standardized tools for geospatial deep learning and geographically structured dataset variants, including the longitude-based EuroSATspatial split [4].

Another line of work has examined the reliability of model evaluation on spatial data. Roberts et al. discussed spatially structured cross-validation, while Kattenborn et al. showed that spatial autocorrelation between training and validation samples can inflate CNN performance estimates [2, 3]. Despite these advances, geographic generalization, multispectral dependence, robustness to missing spectral information, and class-specific performance are commonly evaluated as separate issues rather than within a unified experimental setting.

Structured input dropout has been used to reduce dependence on complete modalities or channels [8], and channel dropout has also been evaluated for multispectral satellite imagery [9]. We adapt this general idea by masking one of four project-defined Sentinel-2 band groups during training.

0.3 Research Questions and Contributions

We study the project around four research questions:

1. How do RGB and 13-band multispectral ResNet-50 models perform under standard and longitude-based spatial evaluation?
2. Can spectral group dropout improve robustness to missing spectral groups, and what clean-performance trade-off

does it introduce?

3. Which EuroSAT classes are most sensitive to input modality and training geography?
4. How do the trained models respond to controlled data corruptions?

Our main contribution is a shared evaluation framework that studies these questions with consistent model families, split manifests, checkpoint selection, and metrics.

1 Dataset and Split Protocol

1.1 EuroSAT Dataset

EuroSAT contains 27,000 Sentinel-2 image patches with a spatial size of 64×64 pixels [1]. The ten classes are Annual Crop, Forest, Herbaceous Vegetation, Highway, Industrial, Pasture, Permanent Crop, Residential, River, and Sea/Lake. Each sample contains 13 spectral bands. The RGB models use B04, B03, and B02, while the MS models use all 13 bands.

1.2 Standard and Spatial Splits

We contrast two evaluation protocols to isolate geographic distribution shifts, and both protocols use fixed 60/20/20 train, validation, and test manifests, corresponding to 16200, 5400, and 5400 images:

1. **Standard Split:** This protocol randomly partitions the dataset into training, validation, and test sets. However, the random splits introduce "information leakage" by assigning geographically adjacent patches to both training and test sets.
2. **Spatial Split:** This protocol ensures the training and testing sets originate from physically distinct regions. By evaluating on imagery from areas unseen, we create a challenging "out-of-distribution" (OOD) scenario. This methodology forces models to rely on class-specific structural and spectral cues that persist across different locations, rather than localized environmental correlations.

This comparison between Random and Spatial protocols measures the robustness of deep learning models in real-world geospatial deployments.

2 Approach

2.1 Architectural Framework

We use ResNet-50 [5] as the shared backbone for all experiments. The RGB model uses ImageNet-pretrained weights [6]. For the 13-band model, we extend the first convolution from three to 13 input channels. The pretrained red, green, and blue filters are copied to B04, B03, and B02. The remaining channels are initialized using the mean of the three RGB filters. The full model is then fine-tuned on EuroSAT.

2.2 Shared Training, Selection, and Evaluation Setup

To ensure consistent comparisons across different modalities and geographic protocols, we established a standardized training and validation pipeline. We apply per-band z-score normalization using the fixed statistics registered for each split protocol. The same statistics are used for training, validation, and test images. This places bands with different numerical scales on comparable ranges.

All models use cross-entropy loss, AdamW, a batch size of 32, seed 42, and validation macro-F1 for checkpoint selection. The RGB baselines use a learning rate of 10^{-3} and weight decay of 10^{-4} , while the MS baselines and dropout runs use a learning rate of 10^{-4} and weight decay of 10^{-3} . Baseline models are trained for 10 epochs, while the dropout sweep uses 20 epochs. Experiment-specific settings for noise augmentation are reported in Section 4.4.

The classification performance was primarily assessed using the Macro-averaged F1 score, which treats all classes equally to account for potential class imbalances. For each class $c \in \{1, \dots, 10\}$, we define precision P_c and recall R_c as:

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c} \quad (1)$$

where TP_c , FP_c , and FN_c denote true positives, false positives, and false negatives. The F1 score for class c is the harmonic mean:

$$F1_c = 2 \cdot \frac{P_c \cdot R_c}{P_c + R_c} \quad (2)$$

The final Macro-F1 metric is the unweighted mean across all $N = 10$ classes:

$$\text{Macro-F1} = \frac{1}{N} \sum_{c=1}^N F1_c \quad (3)$$

All training and evaluation routines were implemented in the PyTorch framework, leveraging TorchGeo to manage complex geospatial data manifests and enforce strict separation between training and evaluation splits, ensuring the integrity of our geographic generalization results.

2.3 Spectral-Group Dropout

To study robustness to missing spectral information, we use spectral-group dropout as training-time regularization. The 13 bands are divided into four groups:

- **Visible:** B02, B03, and B04.
- **Red-edge/NIR:** B05, B06, B07, B08, and B8A.
- **Atmospheric:** B01, B09, and B10.
- **SWIR:** B11 and B12.

After z-score normalization, each training sample is selected for masking with probability p . If selected, one of the four groups is chosen uniformly and all channels in that group are set to zero. A zero after normalization represents

the reference mean for that channel. We do not rescale the remaining channels because the operation simulates missing information rather than ordinary activation dropout.

The operation is enabled only during training. Validation and test inference are deterministic unless a fixed test mask is explicitly requested. This method is intended to encourage the model to rely less on a single spectral group. It does not guarantee complete invariance to missing bands.

2.4 Class-Level Spatial Cross-Evaluation

The random and spatial protocols partition the same EuroSAT sample pool, so direct cross-protocol evaluation may include test samples seen in the alternate protocol’s training or validation split. To avoid this leakage, we construct a shared clean test subset by removing such overlaps from both test sets. All four checkpoints (RGB standard, RGB spatial, MS standard, and MS spatial) are evaluated on this identical subset. For each checkpoint, we compute per-class F1 over the ten EuroSAT classes and macro-F1 as their unweighted mean. This controls test-set variation and isolates the effects of input modality and training split.

3 Experiments and Results

3.1 Baseline Geographic Generalization

The first experiment trains four baseline configurations: RGB standard, RGB spatial, multispectral standard, and multispectral spatial. Each model is selected using validation macro-F1 and then evaluated on the test set belonging to its protocol. This experiment compares both the input modality and the evaluation split while using a shared backbone and checkpoint-selection rule.

Table 1: Generalization Gaps across Modalities

Model	Split	Accuracy	Macro-F1
RGB Baseline	Random	0.9539	0.9529
RGB Baseline	Spatial	0.9220	0.8916
13-Band MS	Random	0.9787	0.9782
13-Band MS	Spatial	0.9469	0.9160

Table 1 shows a consistent gap between the standard and spatial evaluation protocols. Moving from the standard to the spatial split reduces accuracy by 3.19 percentage points for RGB and 3.18 points for MS. The macro-F1 drops are larger: 6.13 points for RGB and 6.22 points for MS. This suggests that the spatial shift affects some classes more strongly than others.

Multispectral input improves macro-F1 over RGB by 2.53 points on the standard split and 2.44 points on the spatial split. Thus, the additional bands provide a similar overall benefit under both protocols, but they do not remove the spatial evaluation gap. Because the two protocols use different test samples, these values describe an evaluation-protocol gap rather

than a controlled causal effect of geography. Section 3.3 provides a same-sample class-level comparison.

3.2 Spectral-Group Dropout Robustness

We apply the spectral-group dropout defined in Section 2.3 while training the 13-band multispectral model on the spatial split. We train with $p \in \{0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$. All runs use the same model, data split, optimizer settings, seed, and 20-epoch budget; only p is changed.

For each run, we keep the checkpoint with the highest validation macro-F1. We then select one value of p using validation performance before evaluating it on the spatial test set. This keeps the test set separate from hyperparameter selection. The selected model and base model are finally evaluated on clean inputs and four fixed spectral-group masks.

Figure 1 shows the validation sweep. The best score is 0.9498 at $p = 0.5$ and epoch 19. The curve is not fully monotonic, but it shows a clear middle-range peak. Too much masking likely removes useful information too often. We therefore freeze $p = 0.5$ for the test comparison.

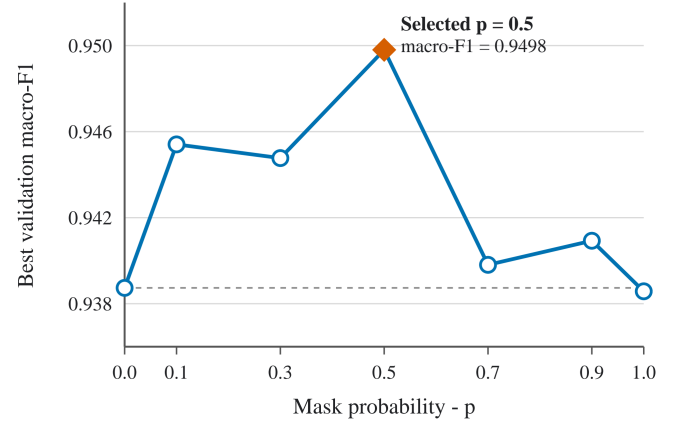


Figure 1: **Validation selection of dropout probability.** Best spatial-validation macro-F1 for each 20-epoch run. The peak at $p = 0.5$ is selected before spatial-test evaluation; $p = 0$ is the control.

We compare the no-dropout control and the selected $p = 0.5$ model without retraining on the same 5400-image spatial test set. Both checkpoints are evaluated on clean inputs and four fixed-mask conditions. Each mask zeros one full spectral group after normalization.

Table 2 shows a clear trade-off. On clean inputs, the selected model loses 1.99 percentage points in macro-F1, but it gains 31.56 points under visible masking. Red-edge/NIR masking gives a smaller gain of 4.27 points. In contrast, the selected model is slightly worse under atmospheric and SWIR masking. These results show partial, condition-specific robustness rather than a general improvement for every missing spectral group.

One possible explanation for the clean-score drop is that the $p = 0.5$ model learns from a mix of complete and incomplete inputs. This may reduce its focus on the strongest

Table 2: Spatial-test macro-F1 under clean and fixed-group masking.

Condition	Base	$p=0.5$	Δ
Clean	0.9160	0.8961	-1.99
Visible masked	0.5226	0.8381	+31.56
Red-edge/NIR masked	0.8335	0.8761	+4.27
Atmospheric masked	0.9172	0.9049	-1.23
SWIR masked	0.9007	0.8695	-3.13

$\Delta = p_{0.5} - \text{baseline}$.

clean spectral combinations. The large visible-mask gain may also be linked to pretraining. The ImageNet RGB filters are mapped to the visible bands, so the control can learn visible features very early. Dropout encourages the model to use non-visible information when those bands are missing.

The SWIR result suggests that dropout may move spectral dependence instead of removing it. SWIR is still available when the visible group is masked, so it may become an important backup signal. This can help under visible masking but hurt when SWIR itself is removed. This explanation is a hypothesis, not a causal result. We also use only one training seed, so we do not report confidence intervals or statistical significance. Fixed zero masking is a controlled test of missing spectral information and does not cover every real sensor failure or image corruption.

3.3 Class-Level Spatial Cross-Evaluation

Figure 2 compares the four checkpoints on the shared overlap-controlled test subset. The largest training-protocol differences were concentrated in agricultural classes. Permanent Crop showed the most consistent degradation under spatial training, decreasing from 0.930 to 0.785 for RGB and from 0.937 to 0.750 for MS. Pasture was also difficult, especially for the MS spatial checkpoint, whose F1 decreased from 0.789 to 0.647. Annual Crop and Herbaceous Vegetation showed smaller but consistent declines under spatial training.

In contrast, Forest, Highway, Industrial, Residential, and Sea/Lake remained comparatively stable across checkpoints. Sea/Lake achieved F1 scores above 0.98 in all settings, while Residential and Forest also remained high. These classes may rely on more geographically stable structural or spectral cues, whereas agricultural classes vary more strongly with crop type, phenological stage, field geometry, soil background, and regional climate.

The effect of multispectral input was also class-dependent. Under spatial training, MS substantially improved River F1 from 0.908 to 0.977 and slightly improved Herbaceous Vegetation, Residential, and Sea/Lake. However, it reduced F1 for Pasture and Permanent Crop. Thus, additional spectral bands benefited selected water- and vegetation-related categories but did not consistently improve the most geographically sensitive agricultural classes.

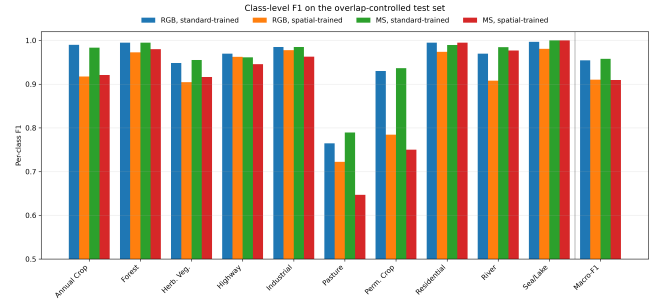


Figure 2: **Class-level F1 on the overlap-controlled common test subset.** All checkpoints are evaluated on identical unseen samples. The largest training-protocol differences occur in agricultural classes, particularly Permanent Crop and Pasture.

These results suggest that geographic sensitivity is concentrated in agricultural classes. Regional differences in phenology, soil, and field geometry are possible explanations, but testing them would require additional metadata or targeted ablations.

3.4 Noise Robustness

Robustness to common input corruptions is an important complement to clean-test accuracy [7]. We evaluate the 13-band spatial model’s sensitivity to two physically motivated input corruptions, and report in the *perturbation ratio* $\langle |\Delta x| \rangle / \langle |x| \rangle$ to compare them on a common axis:

- **Gaussian noise.** Additive $\mathcal{N}(0, \sigma s_b)$, where s_b is the per-band standard deviation from the training set. Scaling by s_b ensures equal relative perturbation across bands.
- **Shot (Poisson) noise.** Each pixel is redrawn as $\text{Poisson}(x/g) \cdot g$, where g is a gain parameter. Variance tracks the signal ($\sqrt{g/x}$ relative noise), so dim bands are perturbed more strongly—the physically correct model for photon-counting sensors.

The experiment has two phases. *Phase 1* evaluates the clean-trained baseline checkpoint across a sweep of noise levels without retraining. *Phase 2* retrains the model with noise-augmented training data, and test the model on the original dataset. All reported numbers are spatial-test macro-F1.

3.4.1 Phase 1: Clean-Model Degradation

Figure 3 (solid lines) shows the degradation of the clean-trained model. At low perturbation the model is stable: Gaussian $\sigma=0.01$ (0.3% perturbation) reduces macro-F1 by only 0.5 points. However, performance collapses at moderate levels. Gaussian $\sigma=0.1$ (2.9% perturbation) drops macro-F1 from 0.898 to 0.534, while Shot noise with $g=5$ (4.0% perturbation) collapses to 0.549. At the strongest levels (Gaussian $\sigma=0.25$, shot $g=10$) the model falls below 0.40. This indicates that the original model cannot survive in the corrupted dataset, and we try to improve this by training on the noise-augmented dataset.

3.4.2 Phase 2: Noise-Augmented Training

We retrain the model with four noise configurations applied only to the training set: Gaussian $\sigma \in \{0.05, 0.1\}$ and shot $g \in \{0.1, 1.0\}$. Each model is then swept across all noise levels on the test set. Table 3 summarizes key operating points.

Table 3: Spatial-test macro-F1 under noise. Each row trains with the indicated noise, and each column shows a evaluation condition.

Training	Clean	G 0.05	G 0.1	S 1
None (baseline)	.898	.754	.534	.725
Gaussian 0.05	.877	.894	.815	.885
Gaussian 0.1	.870	.854	.849	.908
Shot 0.1	.864	.833	.666	.822
Shot 1	.901	.908	.866	.912

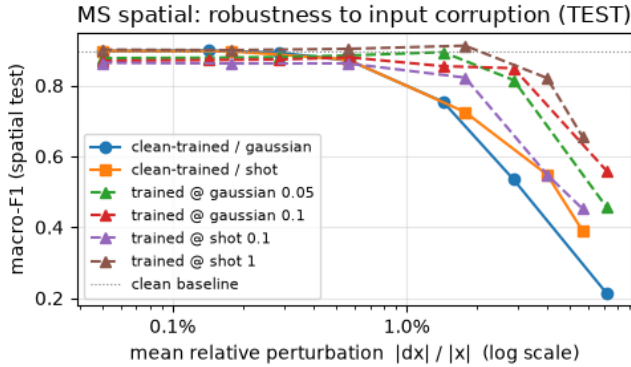


Figure 3: **Noise robustness on spatial test.** Solid lines: clean-trained baseline evaluated under increasing Gaussian and shot noise. Dashed lines: noise-augmented models.

Figure 3 shows that noise-augmented training largely recovers performance, with robustness transferring across noise types (e.g. shot $g=1$ improves Gaussian $\sigma=0.1$ from 0.534 to 0.866). Shot $g=1$ loses only 0.4 points on clean data while improving every tested corruption level; mild corruption (shot $g=0.1$) provides little benefit. Section 5 discusses these patterns further.

4 Discussion

4.1 Augmentation as Regularization

Both spectral-group dropout and noise augmentation improve robustness by injecting structured variation during training, but they differ in an important way. The selected $p = 0.5$ spectral-dropout model loses 1.99 percentage points on clean inputs, whereas shot $g=1$ noise augmentation improves clean test F1 by 0.37 points. One possible explanation is that pixel-level noise acts as a smoother regularizer than zeroing entire band groups, discouraging reliance on high-frequency patterns without removing whole information

channels. Whether the two augmentation strategies provide complementary benefits remains an open question.

The cross-noise transfer result, where shot-trained models improve Gaussian robustness and vice versa, suggests the network learns general perturbation-invariant features. This raises the question of whether a single mixed-noise augmentation policy could replace the per-type sweeps we conducted, and whether the learned invariance extends to corruption types not tested here, such as blur, compression artifacts, or atmospheric distortion.

4.2 Geographic Sensitivity and Class Structure

The concentration of geographic sensitivity in agricultural classes (Permanent Crop, Pasture) likely reflects regional variation in phenology, soil, and field geometry—factors that change across geography more than the structural cues that define classes like Forest or Sea/Lake. However, we did not test this directly. Feature probing or gradient attribution on the spatial vs. standard checkpoints could reveal whether the model relies on texture features that are geographically local. Similarly, layer-level ablations (e.g. freezing the backbone and retraining only the classifier on spatially separated data) could isolate whether the generalization gap originates in learned features or in the final classification layer.

Multispectral input improved some geographically sensitive classes (e.g. River) but hurt others (e.g. Pasture). Understanding when extra bands help and when they introduce spurious regional correlations would benefit from per-band attribution analysis or selective band ablation under the spatial protocol.

4.3 Limitations and Future Work

All experiments use a single architecture (ResNet-50), one dataset (EuroSAT), one spatial split axis (longitude), and one training seed. The generality of these findings to other architectures (e.g. Vision Transformers), datasets with different geographic coverage, or alternative spatial partitioning strategies (e.g. latitude, climate zones) is untested.

Our noise evaluation covers two corruption types at inference time. Real-world sensor degradation includes atmospheric effects, cloud contamination, and radiometric drift, which have different spectral signatures than the synthetic noise studied here. Evaluating on benchmark corruption suites such as those proposed by Hendrycks and Dietterich [7] would provide a broader robustness profile.

Finally, the spectral dropout and noise augmentation experiments were conducted independently. A natural next step is to study their interaction—whether joint augmentation provides additive robustness or whether the two strategies interfere—and to explore adaptive augmentation schedules that adjust dropout probability or noise level during training.

5 Conclusion

This study evaluates geographic, spectral, and noise robustness in EuroSAT land-cover classification using ResNet-50 models with RGB and 13-band multispectral inputs. We discovered that geographic sensitivity is class-dependent: agricultural classes degrade substantially under spatial evaluation while structural classes remain stable, and multispectral input helps selectively rather than universally. Additionally, spectral-group dropout and noise augmentation both improve robustness through training-time variation. Furthermore, noise augmentation avoids the clean-performance penalty that spectral dropout incurs, and its benefits transfer across corruption types. However, we have not tested these robustness interact, for example combining augmentation strategies. Probing the learned features that drive the geographic gap are promising directions for future work. Overall, a single random-split accuracy score is insufficient to characterize model robustness; geographic separation, spectral stress tests, noise sweeps, and class-level analysis together provide a more complete picture.

6 Team Contributions

Member	Contribution Summary
An Xu	Model factory, class-level spatial cross-evaluation.
Qihan Liu	Dataset manifesting, spatial split reader logic.
Shuang Chen	Evaluation harness, JSON artifacts, report structure.
Xufan Lin	Multispectral baselines, spectral-group dropout study, project planning and coordination.

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